

On the Extremal Values of the Weighted Integrity of a Graph

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Abstract

The integrity of a graph G is defined as the minimum value of $|S| + m(G - S)$ taken over all $S \subseteq V(G)$, where $m(H)$ denotes the maximum cardinality of a component of graph H . In this note, we investigate bounds on the maximum and minimum values of the weighted version of this parameter. We also consider the same question for the related parameter vertex-neighbor-integrity.

1 Introduction

There are many ways one can quantify the goodness of a graph as a network. Several of these focus on node failure and the worse case thereof; the associated parameters are sometimes called measures of vulnerability. Examples include connectivity and variants thereof, toughness, integrity, tenacity, and scattering number. More recently, people have considered the generalization of these measures to the case where not all vertices in the graph are considered equally. For example, in many real-world applications, some vertices may be more important or harder to disable than others. In this vein, the concept of weighted integrity was introduced [10] (definitions below) while a weighted version of toughness is given in [8, 12] and a weighted version of scattering number in [9].

We consider here bounds and properties of weighted integrity as well as the weighted version of the related parameter vertex-neighbor integrity. We recall that the integrity of a graph was introduced by Barefoot, Entringer, and Swart [2] in 1987. For any graph G , the *integrity* of G , denoted $I(G)$, is defined as

$$I(G) := \min_{S \subseteq V(G)} \left\{ |S| + m(G - S) \right\},$$

where $m(H)$ is the maximum cardinality of a component of graph H (and 0 if H has no vertex). This parameter was investigated in [6, 7] and elsewhere; for an early survey see [1]. The weighted version of integrity was introduced by Ray and Deogun [10] in 1994. Let G be a graph with weighting w where the weight of every vertex is nonnegative. Then the

weighted integrity of G , denoted as $I_w(G)$, is

$$I_w(G) := \min_{S \subseteq V(G)} \{w(S) + m_w(G - S)\},$$

where $m_w(X)$ denotes the weight of the component of graph X that has largest weight, and $w(X)$ denotes the sum of the weights of vertices in X . We define an *I-set* as any subset S that achieves the minimum. In particular, Ray and Deogun showed that calculating the weighted integrity of a tree in general is NP-complete. Further, Ray et al. [11] provided a polynomial-time algorithm for interval graphs, which includes paths. More discussion of the complexity is given in [4].

In this note we provide some basic properties of weighted integrity, focusing on bounds for the minimum and maximum weighted integrity of a general graph. Thereafter we consider some similar questions for the weighted version of vertex-neighbor-integrity that was introduced by Cozzens and Wu [3] in 1996.

2 Bounds on Weighted Integrity

In order to compare integrity and weighted integrity fairly, we define a *normalized weighting* as one where the total weight of the vertices sums to the order of the graph. Note this means that when all weights are 1, we recover the ordinary integrity of the graph exactly. Also note that one can normalize any given weighting of a graph.

We first investigate the weighted integrity of the two most extreme cases: when G is edgeless, and when G is complete.

Proposition 1. *Let G be a graph with normalized weighting w . Then $I_w(G) \geq 1$ with equality if and only if G is edgeless and every vertex has weight 1.*

Proof. Suppose first that G is edgeless with every vertex having weight 1. Then $I_w(G) = 1$, because we can remove no vertices and all components have weight 1. Now suppose that $I_w(G) = 1$. If there is a vertex that has weight greater than 1, then either S has weight more than 1, or there is a component of $G - S$ of weight greater than 1 and thus $I_w(G) > 1$. So all vertices must have weight 1. But then if there are two vertices that are adjacent, either $w(S) \geq 1$ and $m_w(G - S) \geq 1$, or there is a component of weight greater than 1. It follows that G is edgeless. $\square$

At the other extreme:

Proposition 2. *For any normalized weighting w , $I_w(K_n) = n$.*

Proof. If S is any subset of the vertices, it is immediate that $K_n - S$ is connected (or vertexless) and thus $m_w(K_n - S) = n - w(S)$. It follows that $I_w(K_n) = w(S) + n - w(S) = n$. $\square$

It is easy to see that if G is not complete, then we may assume S is a cut-set. But Ray et al. [11] went further and define a set as *strong* if $G - S'$ has fewer components than $G - S$ for all S' strictly contained in S .

Lemma 3. [11] *For any graph G and weighting w , there exists an I -set that is strong.*

We now consider the complete bipartite graph:

Proposition 4. *Let $K_{r,s}$ be the complete bipartite graph with weighting w and let $x_1, \dots, x_r$ represent the weights of the vertices in the first partite set and $y_1, \dots, y_s$ the weights of the vertices in the second partite set. Then*

$$I_w(K_{r,s}) = \min \left\{ \left(\sum_{i=1}^r x_i \right) + \max_{1 \leq j \leq s} \{y_j\}, \left(\sum_{j=1}^s y_j \right) + \max_{1 \leq i \leq r} \{x_i\} \right\}.$$

Proof. The strong cut-sets are the two partite sets. If one removes a partite set, then the remaining component of largest weight is just the vertex of largest weight from the other partite set. So the weighted integrity is the minimum of the two values obtained from removing each set, hence the result. $\square$

We next consider the maximum possible value of the weighted integrity for a given graph:

Proposition 5. *Let G be a graph of order n . Then for any normalized weighting w of G , $I_w(G) \leq n$. Furthermore, there exists a normalized weighting w such that $I_w(G) = n$.*

Proof. Trivially $I_w(G) \leq n$, since the sum of all the weights equals n . To achieve this bound, simply give one vertex u weight n and all other vertices weight 0. Either the vertex u contributes to $w(S)$ or to $m_w(G - S)$; and then $I_w(G) = n$. $\square$

In the previous result, we determined the maximum possible weighted integrity of a graph. We now determine the minimum possible weighted integrity. We need the following lemma.

Lemma 6. *For normalized weightings, the minimal possible weighted integrity of a noncomplete graph G occurs when the I -set S has weight 0 and the rest of the weight is distributed equally amongst the components of $G - S$.*

Proof. Assume $I_w(G)$ is minimized over all normalized weightings w of G with S the I -set. Since G is noncomplete, the subgraph $G - S$ has at least two components. Suppose $w(S) = x > 0$. Then $I_w(G) = w(S) + m_w(G - S)$. So if we rearrange the weight such that $w(S) = 0$ while reallocating the weight x equally to each component in $V(G) - S$, then the weight of each component increases by at most $x/2$, since there are at least two components. Thus, $I_{w'}(G) < I_w(G)$, a contradiction of the choice of w . That is, $w(S) = 0$. Furthermore, the components that remain should have weight evenly distributed across them. $\square$

We now determine the minimum possible weighted integrity that a general graph may have.

Proposition 7. *Let G be a graph of order n and independence number α . Then for any normalized weighting w of G ,*

$$I_w(G) \geq n/\alpha.$$

Furthermore, there exists a normalized weighting w of G such that $I_w(G) = n/\alpha$.

Proof. By Lemma 6, when w is optimal it holds that

$$I_w(G) = \min_{S \subseteq V(G)} \left\{ w(S) + \frac{n - w(S)}{k(G - S)} \right\} = \min_{S \subseteq V(G)} \left\{ \frac{n}{k(G - S)} \right\}$$

where $k(G - S)$ is the number of components remaining after removing S . This quantity is minimized when the number of components in $G - S$ is maximized. But the maximum number of components that can occur when vertices are removed is equal to α . Thus $k(G - S) \leq \alpha$ and so $I_w(G) \geq \frac{n}{\alpha}$.

To show the tightness of this bound, let C be a minimum vertex cover of G and let every vertex of C have weight 0. Recall that the removal of a minimum vertex cover leaves a maximum independent set. So if we assign each of the α vertices in $G - C$ the weight n/α , then $I_w(G) = n/\alpha$. $\square$

Proposition 7 provides the general lower bound for a graph G , which gives rise to the following corollaries.

Corollary 8. *For a normalized weighting w , the weighted integrity of a connected graph G of order n is at least $n/(n-1)$, with equality if and only if $G = K_{1,n-1}$ where the vertex of degree $n-1$ has weight 0 and all other vertices have weight $n/(n-1)$.*

Proof. A connected graph has independence number at most $n-1$. So the bound follows from Proposition 7. To achieve the bound, the graph must have $\alpha = n-1$ and so must be a star. $\square$

Similarly, the weighted integrity of a k -connected graph of order n is at least $n/(n-k)$.

Corollary 9. *For a normalized weighting, the weighted integrity of the complete bipartite graph $K_{r,s}$ where $r \leq s$ is at least $(r+s)/s$.*

Proof. Notice that $\alpha = s$ and apply Proposition 7. $\square$

Recall that for ordinary integrity the chromatic number of G is a lower bound for $I(G)$ (see [1]). But the above proposition shows that this is not the case for weighted integrity, as for example, the star can have weighted integrity smaller than 2 for a normalized weighting.

We now show that for a general graph G it is possible for $I_w(G)$ to take on all values between the bounds of Propositions 7 and 5.

Proposition 10. *Let G be a graph of order n and independence number α . Then there exists a normalized weighting w_x of G such that $I_{w_x}(G) = x$ whenever $\frac{n}{\alpha} \leq x \leq n$.*

Proof. Define the weighting w_x of G where some minimum vertex cover C is given total weight 0, one of the vertices of $G-C$ is given weight x , and all remaining vertices of $G-C$ are given weight $(n-x)/(\alpha-1)$. Note that the total weight is $x + (\alpha-1)(n-x)/(\alpha-1) = n$, as required. Further, since $x \geq n/\alpha$ it follows that $x \geq (n-x)/(\alpha-1)$ and so $m_w(G-C) = x$. Thus $I_{w_x}(G) = x$. $\square$

3 Weighted Integrity and Some Graph Operations

In this section we briefly consider some graph operations. If the weighting of the resultant graph is arbitrary, then the previous results apply; so we assume that the operation also specifies the weighting.

As is the case for integrity, the deletion of an edge or vertex cannot increase the weighted integrity. For the join and the disjoint union, the results mirror those for ordinary integrity, given in [6]. For both operations, we assume the vertices of the combination graph retain the weights they have in the constituent graphs.

Proposition 11. *Let G and H be graphs with weighting w_G and w_H respectively. If W is their disjoint union and X their join, and w the resultant weighting, then*

$$\max\{I_{w_G}(G), I_{w_H}(H)\} \leq I_w(W) \leq I_{w_G}(G) + I_{w_H}(H)$$

and

$$I_w(X) = \min\{I_{w_G}(G) + w_H(V(H)), I_{w_H}(H) + w_G(V(G))\}.$$

Proof. If $S_G \subseteq V(G)$, $S_H \subseteq V(H)$, and $S = S_G \cup S_H$, then $w(S) = w(S_G) + w(S_H)$ and $m_w(W - S) = \max\{m(G - S_G), m(H - S_H)\}$. Thus the bounds on the union follow. For the join X , note that every strong set contains either all of $V(G)$ or all of $V(H)$. $\square$

We next consider the operation of replacing each vertex by a clique. Specifically, we consider the *composition* (or lexicographic product) $G[K_r]$ where every vertex v of G is replaced by a copy H_v of K_r where if v and w are adjacent in G then all vertices of H_v are adjacent to all of H_w .

Proposition 12. *Let G be a graph with weighting w . If Y is the lexicographic product $G[K_r]$ where vertices of Y inherit their weight from G , then*

$$I_w(Y) = rI_w(G).$$

Proof. Let S be an I -set for Y . By Lemma 3, we may assume that $S = S' \times V(K_r)$ for some $S' \subseteq V(G)$. Then $w(S) = r w(S')$ and $m_w(Y - S) = r m_w(G - S')$. Thus the weighted integrity of Y is just $rI_w(G)$. $\square$

Finally in this section we consider a Nordhaus–Gaddum bound. Specifically we establish a result for the sum of the weighted integrities of a graph G and its complement $\overline{G}$ with the same weighting.

Proposition 13. *For a graph G of order n with normalized weighting w ,*

$$I_w(G) + I_w(\overline{G}) > n$$

and this bound is sharp.

Proof. Let S be an I -set of G . Say the subgraph $G - S$ has components $H_1, \dots, H_k$ where H_1 has maximum weight. Thus $I_w(G) = w(S) + w(V(H_1))$. Let T be an I -set of $\overline{G}$ and let $Y = V(G) - S$. There are two cases.

(1) Assume that in $\overline{G}$ the subgraph induced by $Y - T$ is connected. Then

$$I_w(\overline{G}) = w(T) + m_w(\overline{G} - T) \geq w(Y \cap T) + w(Y - T) = w(Y).$$

Thus $I_w(G) + I_w(\overline{G}) \geq n + w(V(H_1))$. For the sum to equal n , it is necessary that $w(V(H_1)) = 0$ and so $w(G - S) = 0$ (since H_1 has maximum weight); hence $I_w(G) = n$. But we know that $I_w(\overline{G}) > 0$ (by Proposition 1), and so the strict inequality follows.

(2) Assume that in $\overline{G}$ the subgraph induced by $Y - T$ is not connected. Then T includes all of $V(H_i)$ for at least $k - 1$ of the H_i . So say T contains all $V(H_i)$ except some subset of $V(H_j)$. Then $I_w(\overline{G})$ is at least $\sum_{i \neq j} w(V(H_i))$ plus the largest weight, call it ϕ , of a vertex in $S \cup V(H_j)$. It follows that $I_w(G) + I_w(\overline{G}) \geq w(S) + w(V(H_1)) + \sum_{i \neq j} w(V(H_i)) + \phi \geq n + \phi$. For the total sum to equal n , it is necessary that (i) all vertices of $S \cup V(H_j)$ have weight 0, and (ii) that $w(V(H_1)) = w(V(H_j))$. This means the total weight is 0, which is a contradiction. $\square$

The above bound can be achieved as follows. Consider the graph G_n that consists of $n/2 - 1$ isolated vertices each of weight 2 and one clique of $n/2 + 1$ vertices of total weight 2 equally distributed. Then $I_w(G_n) = 2$ and $I_w(\overline{G}_n) = 2(n/2 - 1) + 2/(n/2 + 1)$ so that the total integrity sum is $n + 2/(n/2 + 1)$.

4 Vertex-Neighbor Integrity

In this section, we briefly consider the analogous bounds for the weighted version of vertex-neighbor integrity. For a graph G and set $S \subseteq V(G)$, let G/S denote the subgraph $G - N[S]$; that is, one removes S and all of its neighbors. Cozzens and Wu [3] defined the *vertex-neighbor integrity* of G to be

$$\text{VNI}(G) = \min_S |S| + m(G/S),$$

where as before, $m(G/S)$ is the maximum order of a component of G/S , and specifically, it is 0 if G/S is vacuous.

For example, the complete graph has $\text{VNI} = 1$, immediately emphasizing that this parameter is not monotonic. Cozzens and Wu [3] showed that, out of the trees of a given order, the path has maximum value, which is $\lceil 2\sqrt{n+3} \rceil - 4$, for $n \geq 2$. Gambrell [5] observed that $\text{VNI}(G) \leq n/2$ for any connected graph with $n \geq 2$ and showed that there are graphs, namely expanders, where VNI is linear in the order. The exact best possible asymptotics remain unresolved.

One can define a weighted version of this parameter, as was done for weighted integrity. Specifically, if w is a weighting, then we define the *weighted vertex-neighbor integrity* to be

$$\text{VNI}_w(G) = \min_S w(S) + m_w(G/S).$$

Proposition 14. *For an isolate-free graph G of order n and a normalized weighting w of G , it holds that $0 \leq \text{VNI}_w(G) \leq n/2$ and this bound is sharp.*

Proof. The lower bound is immediate. For the upper bound, let S be a maximal independent set of G and let $\bar{S} = V(G) - S$. Then since S is maximal, every vertex in $\bar{S}$ has a neighbor in S ; since there are no isolates, every vertex in S has a neighbor in $\bar{S}$. It follows that both subgraphs G/S and $G/\bar{S}$ have no vertices. Thus $m_w(S) + m_w(\bar{S}) + m_w(G/S) + m_w(G/\bar{S}) = n$, whence the bound by pigeonhole. $\square$

The lower bound is attainable for every isolate-free graph. Indeed, for any graph with an edge, one can put all weight on one vertex v and then choose $S = V(G) - \{v\}$ and so have $\text{VNI}_w(G) = 0$. The upper bound is attainable for trees, as we observe next.

Proposition 15. *If G is a graph of order n and minimum degree 1, then the maximum value of $\text{VNI}_w(G)$ over all normalized weightings w is $n/2$.*

Proof. The upper bound follows from the above proposition. For a weighting w achieving the value, pick one end-vertex x with neighbor y , and give both x and y weight $n/2$ and all other vertices weight 0. Consider any set $S \subseteq V(G)$. If S contains at least one of x or y , then $w(S) \geq n/2$. Otherwise, x in G/S and so $m_w(G/S) \geq n/2$. $\square$

Another example of equality in Proposition 14 is C_4 with the uniform weighting.

5 Future Work

For further research, it would be interesting to what other classes of graphs have polynomial-time algorithms for calculating the weighted integrity and which do not. Also, more calculations and bounds on both parameters for specific families of graphs would be helpful.

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